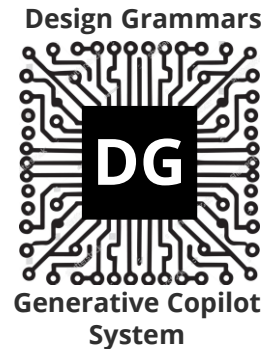


PhD Thesis:

Ontology-based Generative Design System in Architecture integrated with open BIM environment: encoding and incorporating architectural knowledge throughout design lifecycle to enhance integrity in AEC



02 CONCEPT STAGE

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Laboratório de Paisagens,
Património e Território



Fundação
para a Ciência
e a Tecnologia

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ONTOLOGY

Prototype Development

Contents

- 1. Subject mapping**
- 2. Why Ontology is needed**
- 3. Ontology Development Workflow**
 1. DEFINE THE DOMAIN AND SCOPE
 2. ALIGN WITH EXISTING ONTOLOGIES AND REUSE
 3. KEY CONCEPTS LIST
 4. TAXONOMY OF CLASSES (CLASSES & HIERARCHY)
 5. CLASS PROPERTIES – ATTRIBUTES & RELATIONSHIPS
 6. ONTOLOGY MAPPING

2 CONCEPT STAGE. ONTOLOGIES

3 WORKPLAN

Stage	Task #	Tasks	1st Semester						2nd Semester					
			Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug
Exploration Stage	T1.1	Problem Statement. Survey scientific problem regarding information design systems, design knowledge integration and sharing across multidisciplinary environment, automation tools applied to facilitate this process.		✓	✓	✓								
	T1.2	Data-Driven Design. Review data-driven design approach and its implementations.				✓	✓	✓						
	T1.3	Generative Systems. Investigate integration of Generative Systems into design process and assistance in generation, evaluation, and optimization of design states.				✓	✓	✓	✓					
	T1.4	Design Ontologies. Review Design Ontologies for the organisation and connection of concepts from diverse domains of Architectural Knowledge. Infer and draft an ontology prototype.					✓	✓	✓	✓				
	T1.5	Design Databases. Review Design Databases for collecting , storing, and transferring heterogeneous knowledge data throughout design lifecycle.						✓	✓	✓	✓			
	T1.6	Draft State-of-Art Chapter of the dissertation								✓	✓	✓		
Concept Stage	T2.1	Case study analysis and planning: Generative Scheme. Concept case study selection, input data, grammar, and design objectives analysis. Definition of the Generative Scheme of the prototype system and use case analysis.								✓	✓			
	T2.2	Ontology. Creating ontology prototype based on collected concepts in the survey and extending it									✓			
	T2.3	Methodology elaboration. Includes integration of Knowledge system with ontology, design platform, formats, interfaces, and interoperability analysis.									✓			
	T2.4	Documenting results and writing dissertation												
	T2.5	Publication in scientific conference or magazine.												

T1.1: **Problem Statement.** Survey scientific problem regarding information design systems, design knowledge integration and sharing across multidisciplinary environment, automation tools applied to facilitate this process.

T1.2: **Data-Driven Design.** Review data-driven design approach and its implementations.

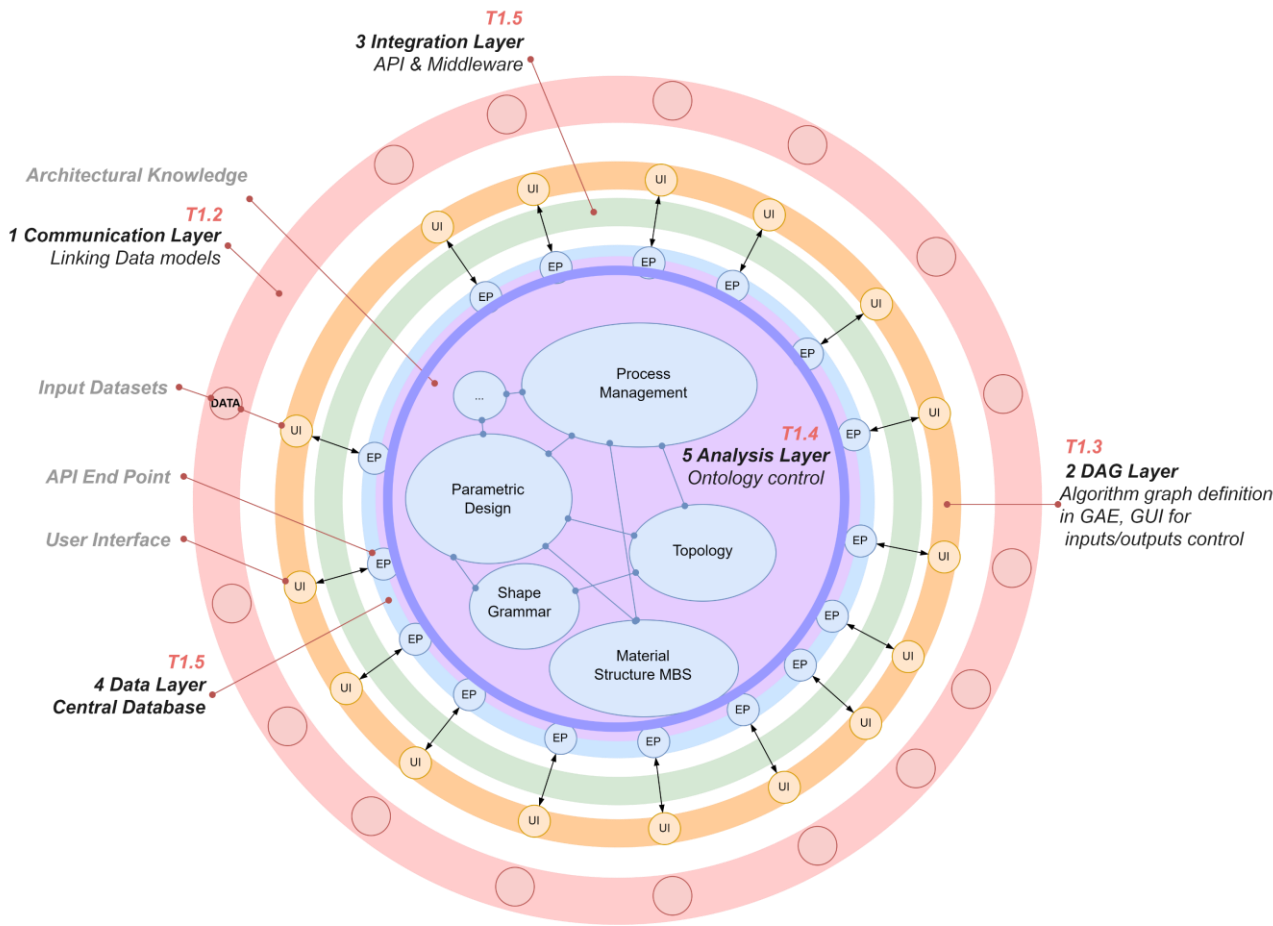
T1.3: **Generative Systems.** Investigate integration of Generative Systems into design process and assistance in generation, evaluation, and optimization of design states.

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T1.5: **Design Databases.** Review Design Databases for collecting , storing, and transferring heterogeneous knowledge data throughout design lifecycle.

T1.6: Draft State-of-Art Chapter of the dissertation

Subjects mapping with the proposed framework



2 CONCEPT STAGE. ONTOLOGIES

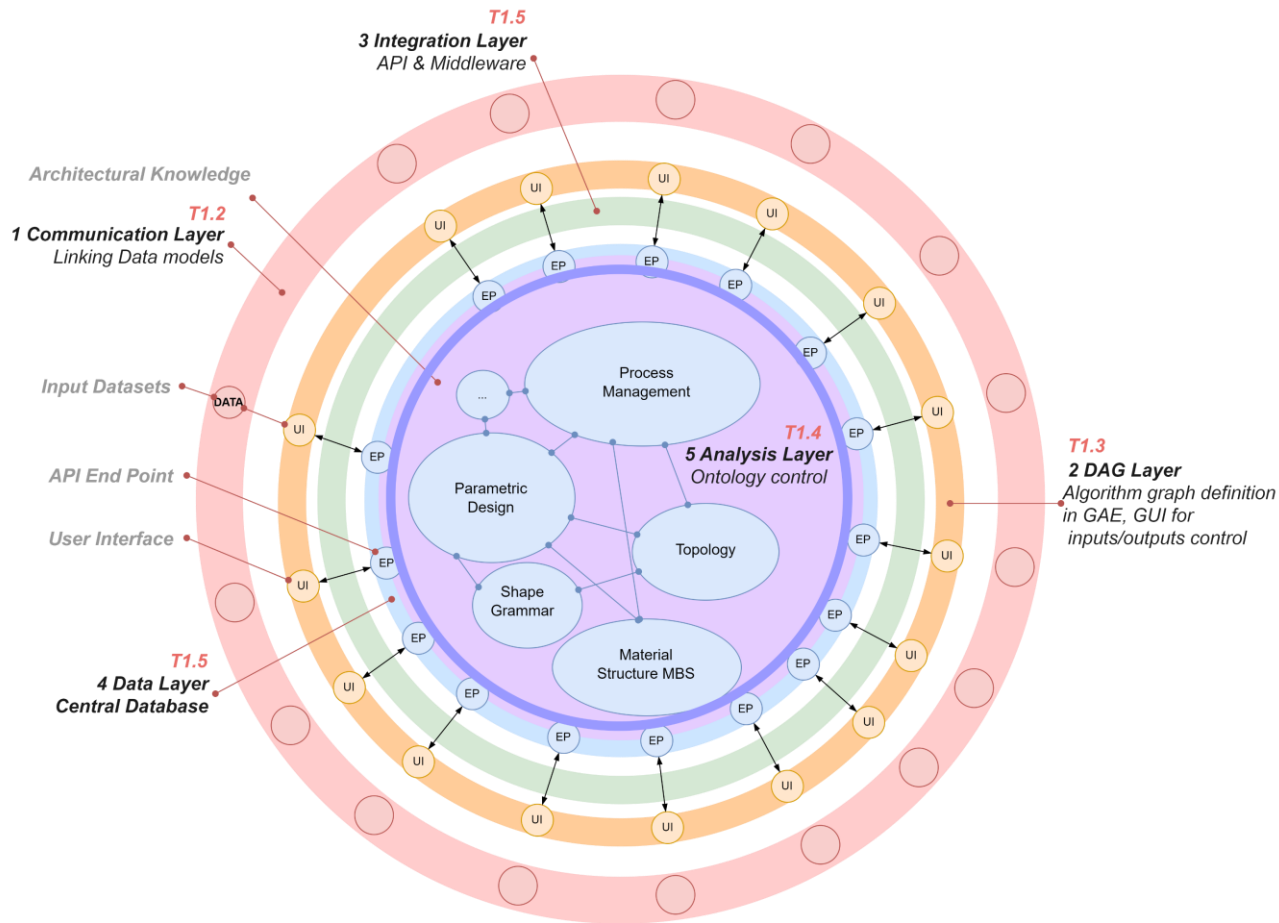
1 SUBJECT MAPPING

4 Ontologies. Ontology is a brain of the Knowledge Graph, it interprets all the entities and deduct semantic data in response to user request to the system.

To organize and connect diverse domains, it is necessary to have a common understanding and shared knowledge of the concepts associated with the design. Thus, elaborating Ontology, which embraces and extends existing design ontologies and their interrelationships, is needed. Ontology will define the organizational semantic core of the system. (101314)

* **SoA** - acronym for State of Art

Subjects mapping with the proposed framework



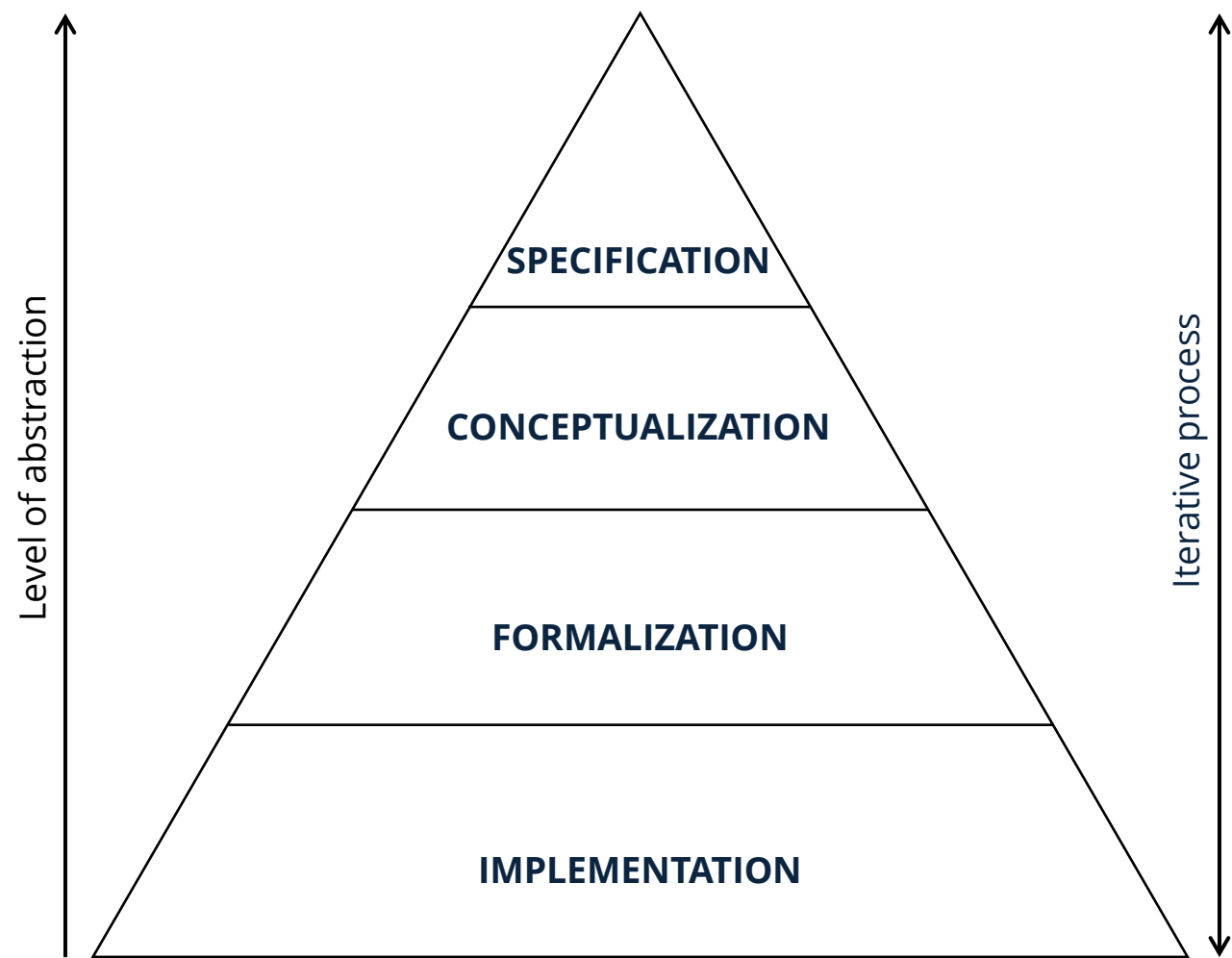
3 ONTOLOGY DEVELOPMENT METHODOLOGY

**“There is no one correct way to model a domain. There are
always viable alternatives.”**

Natalya F. Noy, Stanford University, 2000

3 ONTOLOGY DEVELOPMENT METHODOLOGY

Ontology Development Oriented Activities (Stages). [13]



3 ONTOLOGY DEVELOPMENT METHODOLOGY

Ontology Development Oriented Activities (Stages). [13]

STEP 1 SPECIFICATION.

Main definition of Ontology

- ☐ **1.1** Define the domain and scope.
- ☐ **1.2** Add complimentary specifications

STEP 2 CONCEPTUALISATION.

Structuring domain knowledge in conceptual model

- ☐ **2.1** Key concepts domain (terminology)
- ☐ **2.2** Taxonomy of classes (classes & hierarchy)
- ☐ **2.3** Class properties – attributes & relationships
- ☐ **2.4** Property constraints

STEP 3 FORMALISATION.

Formalize conceptual model in (semi-)computable model

STEP 4 IMPLEMENTATION.

Construction of a computable model in an ontology representation language

Support Activities

- ☐ **Knowledge Acquisition** – gathering knowledge from experts (*Literature Review*)
- ☐ **Evaluation** – technical evaluation of the ontology in each step of the development process
- ☐ **Reuse of existing ontologies, merging and alignment**
- ☐ **Documentation** – each step of the ontology development must be accurately documented
- ☐ **Configuration Management** – version control

STEP 1.1 SPECIFICATION. DEFINE THE DOMAIN AND SCOPE.

DOMAIN – *Procedural Design in Conceptual Stage of Architecture*

SCOPE - *Design Grammars (Problem formulation / Parametric schema, design inputs, constraints, relationships)*

PURPOSE - *Support for design system decomposition and grammar inferring; transition of design intent through later stages of parametric design and grammar validation.*

END-USERS – *Architects, Clients, etc*

MAINTAINERS – *BIM Department*

BENEFITS – *Guarantee rigid communication of design grammars, simplify complex design ideas and multiple constraints management, enable automatic design intent validation, raise sustainability during design reformulations and reuse, etc*

STEP 1.2 SPECIFICATION. ADD COMPLIMENTARY SPECIFICATIONS

a. Tasks – *list tasks/problems to solve, their arrangement (composition, order), and required resources to fulfil the tasks*

b. Resources – *Quality assurance of all produced **resources** during development (ontologies, software, documentation)*

c. Environment – *Designated software platform for the ontology, and applications that use the ontology*

d. Feasibility – *answer the questions: Can the ontology be developed? Does it make sense to develop it? What Competency Questions should be answered by the Knowledge represented in the ontology*

2 CONCEPT STAGE. ONTOLOGIES

3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 1.2a TASKS

list tasks/problems to solve with Ontology, their arrangement (composition, order), and required resources to fulfil the tasks

- a. **Tasks**
- b. *Resources*
- c. *Environment Study*
- d. *Feasibility Study*

HOW DO ARCHITECTS (DESIGNERS) THINK:

- a. Problem–solution **coevolution** [2,3,4]
- b. **Iterative** divergence and convergence of the problem and solution space and problem **reformulation** [5,6,10]
- c. Critical role of framing and **abductive reasoning** in the design process [11]
- d. **Analogical design** reasoning. Design-by-Analogy (DbA), Case-Based Reasoning (CBR) [7, 8]
- e. **Function-structure** mapping and decomposition [9]

TASKS SOLVED BY ONTOLOGY:

- a. **Decompose** Design Problem and encode into **computable** parts
- b. Enable decomposed parts to **generate solution space** by **iterative** interaction
- c. Enable **modular** structure of Design Knowledge
- d. Compose complex algorithms (ideas) within a **single class and its properties**
- e. Raise Design System sustainability during **reformulations** due to **decentralized composed** structure
- f. Precisely map Design problem with **Analogies** and **evaluate**
- g. Raise Design Knowledge **Interoperability and Perception** for all stakeholders
- h. **Translate** Design Knowledge among authoring systems
- i. Enable user to build **Semantic reasoner** with defined **design rules**
- j. Enable automatic **design intent validation**
- k. Enable **AI Models integration** for inferring, improving or enriching design knowledge

2 CONCEPT STAGE. ONTOLOGIES

3 ONTOLOGY DEVELOPMENT WORKFLOW

- a. Tasks
- b. Resources**
- c. Environment Study
- d. Feasibility Study

STEP 1.2b RESOURCES

Quality assurance of all produced resources during development (ontologies, software, documentation)

Syntax: RDF, SWRL, OWL, DATALOG

Software: Proteje

Ontologies: FBS, DCM, Topologic, BOT, SWRL

- ☐ Design Knowledge Ontology (**Gero FBS**, SAPPhIRE, BFO)
- ☐ Topologic Ontology (**BOT**)
- ☐ Parametric Design Ontology (**DCM**, AMV)
<https://www.isibang.ac.in/ns/amv/index.html#http://purl.org/dc/terms/>
- ☐ Shape Grammar Ontology
- ☐ Building Product Ontology <https://github.com/w3c-lbd-cg/product>

Use/Re-Use analysis: versatility of implementation, accessibility of ontology for external usage and integrations, reuse of existing ontologies

Maintenance analysis: ability to update and adjust the ontology

2 CONCEPT STAGE. ONTOLOGIES

3 ONTOLOGY DEVELOPMENT WORKFLOW

There isn't a single, universally accepted FBS ontology, but several researchers have published versions. One widely cited version is by Gero and Kannengiesser

STEP 1.2b : RESOURCES

Design Knowledge Ontology
Gero's FBS: **FUNCTION – BEHAVIOUR - STRUCTURE**

FBS Design reasoning framework

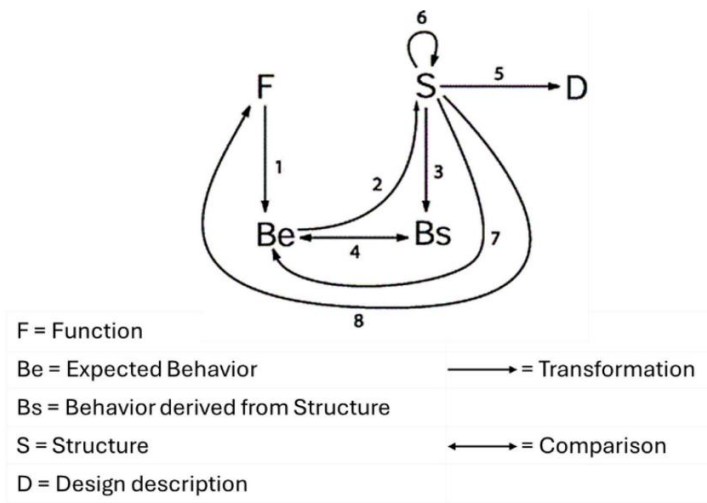


Fig. 3. Gero's design reasoning framework using FBS ontology[29].

Example: Chair Design

Designing a chair decomposes into sub-functions (e.g., "support weight") mapped to structures (e.g., legs), with behavior (e.g., load capacity) mediating the relationship. This approach supports modular, transparent, and domain-agnostic design reasoning.

Function seating	Expected behaviors (Be) load capacity, comfort of seating, lightweight	Structure solution alternatives
Actual behavior (Bs) Actual load capacity, comfort of seating, lightweight		
Design Description.	Reformulation Generate another solutions Bs Change expectations in Be Change target function F	

3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 1.2b : RESOURCES

Design Knowledge Ontology

Design Criteria Modelling (DCM)

Stage of Development: **Conceptualization**

Domain: **Conceptual Design Project**

3 types of design information in DCM: **Semantic** + **topological** + **geometric**

Manually developed **Design Criteria** are transformed into **Semantic Ontology**

Semantic Web Rule Language (SWRL) expression is used to associate **Semantic Ontology** with topological relationships:

Antecedent(Body) $\rightarrow$ Consequent (Head)

Antecedent(Body) = $\text{atom_1}(\text{?x}) \wedge \text{atom_2}(\text{?y}) \wedge \text{atom_N}(\text{?x}, \text{?y})$

Topological relationships are computed algorithmically

Basic Topological relationships: 'separated', 'connective', 'adjacent', 'enclosing'

Geometric objects are used **as input and output parameters** of topological algorithms associated with Semantic Ontology

Example:

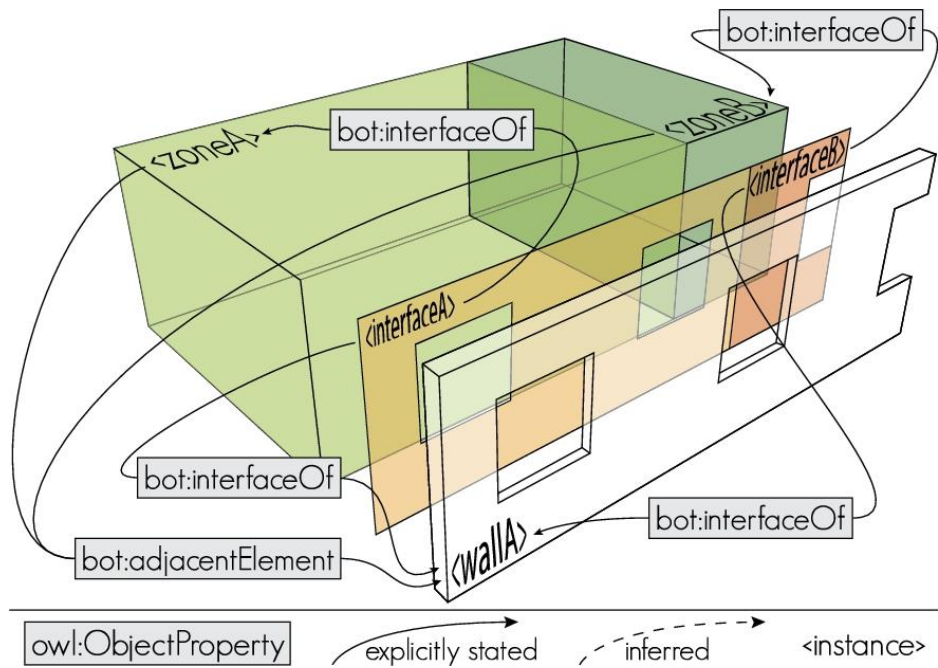
$\text{IndoorSpace}(\text{?s}) \wedge \text{Window}(\text{?w}) \wedge \text{hasWindow}(\text{?s}, \text{?w}) \wedge \text{OutdoorSpace}(\text{?x})$
 $\wedge \text{LandscapeElement}(\text{?y}) \wedge \text{hasQuality}(\text{?y}, \text{"Good"}) \wedge \text{isAdjacent}(\text{?s}, \text{?x}) \wedge \text{FaceTo}(\text{?w}, \text{?y})$
 $\rightarrow \text{hasVisionQuality}(\text{?s}, \text{"Good"})$

STEP 1.2b : RESOURCES

Topologic Ontology

BOT: BUILDING TOPOLOGY ONTOLOGY

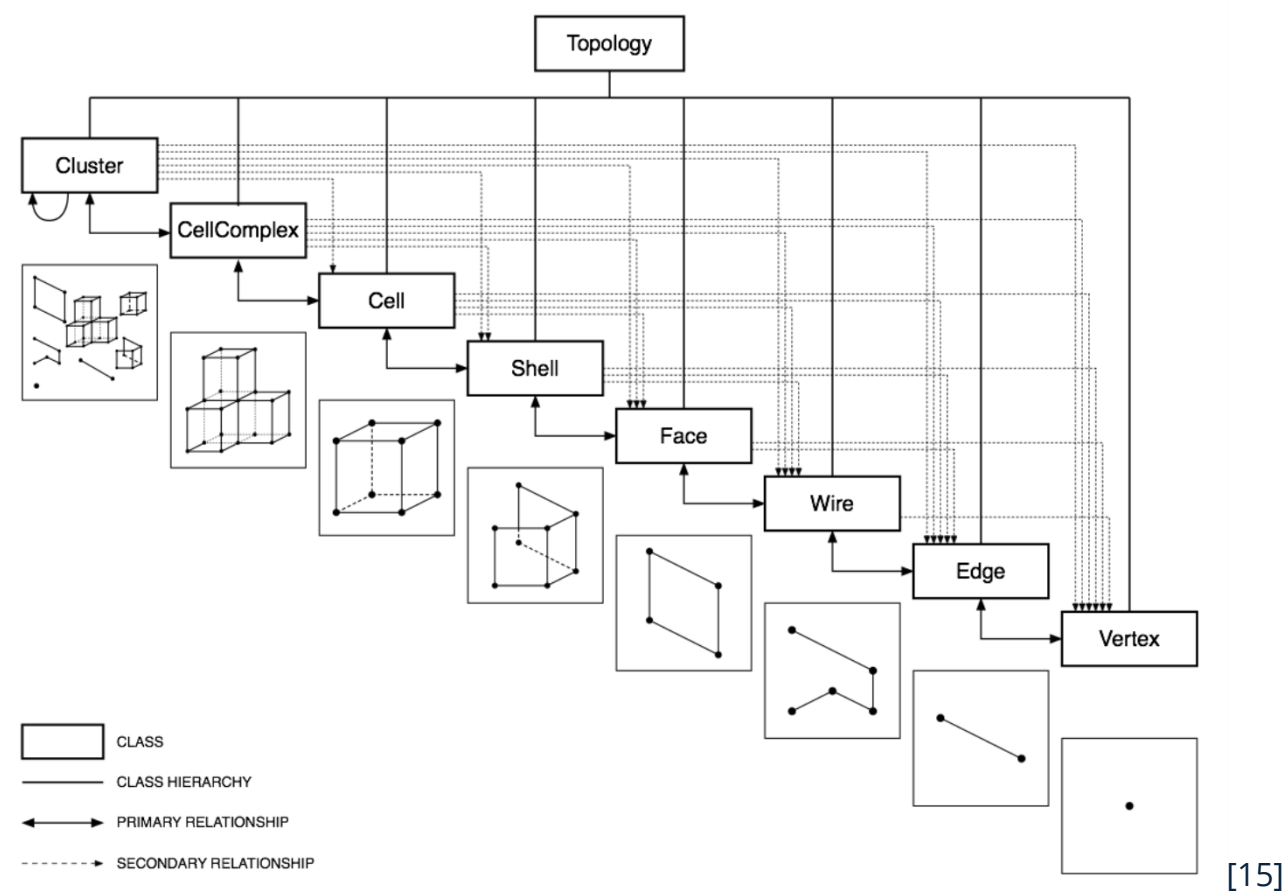
<https://w3c-lbd-cg.github.io/bot/>



3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 1.2b : RESOURCES

Topologic Ontology
TOPOLOGIC Ontology



- a. Tasks
- b. Resources
- c. **Environment Study**
- d. Feasibility Study

STEP 1.2c : ENVIRONMENT

Designated software platform for the ontology, and applications that use the ontology

BIM-modelling and Parametric Design software, Visual Programming Editors

Applications: Rhinoceros, Revit

Visual Programming Editors: Grasshopper, Dynamo

Database: Neo4j, ...

Development Software: Proteje

- a. Tasks
- b. Resources
- c. Environment Study
- d. Feasibility Study

By employing algorithmic components in Grasshopper, simple topologies between two geometries are more easily detected than before. However, as composing algorithms within Grasshopper is usually too complex to be easily recognized by most architects, **composing complex or multiple algorithms within a single topological class** is usually more convenient for most users. [12]

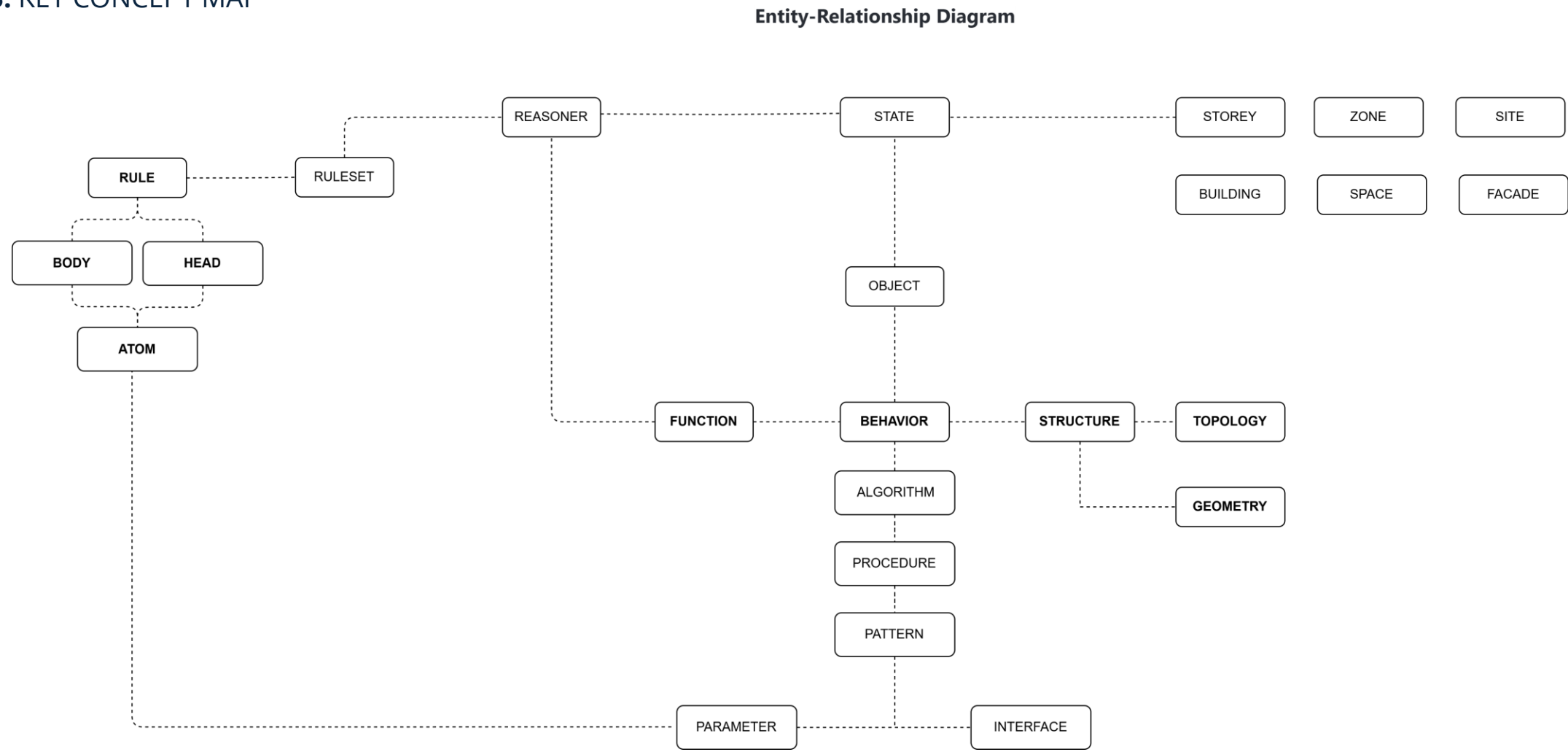
STEP 1.2d : FEASIBILITY

COMPETENCY QUESTIONS / REQUESTS TO SYSTEM

- a. Does the staircase comply with the defined dimension constraints and spatial adjacencies?
- b. Does the current model state meet the lighting constraints?
- c. Generate design space of possible plan configurations according to the spatial program and constraints defined.
- d. Extract QTO based on defined categories list and calculation methods.
- e. Find all soft clashes in the model based on the defined clash schema.
- f. Does the furnishing & equipment layout comply with the accessibility requirements defined in the linked codes?
- g. Does the BIM-Model at the Detailed Design Stage comply with the initial parametric schema at the Conceptual Design Stage?
- h. Does current structural column layout complies with façade design criteria requirements?

3 ONTOLOGY DEVELOPMENT WORKFLOW

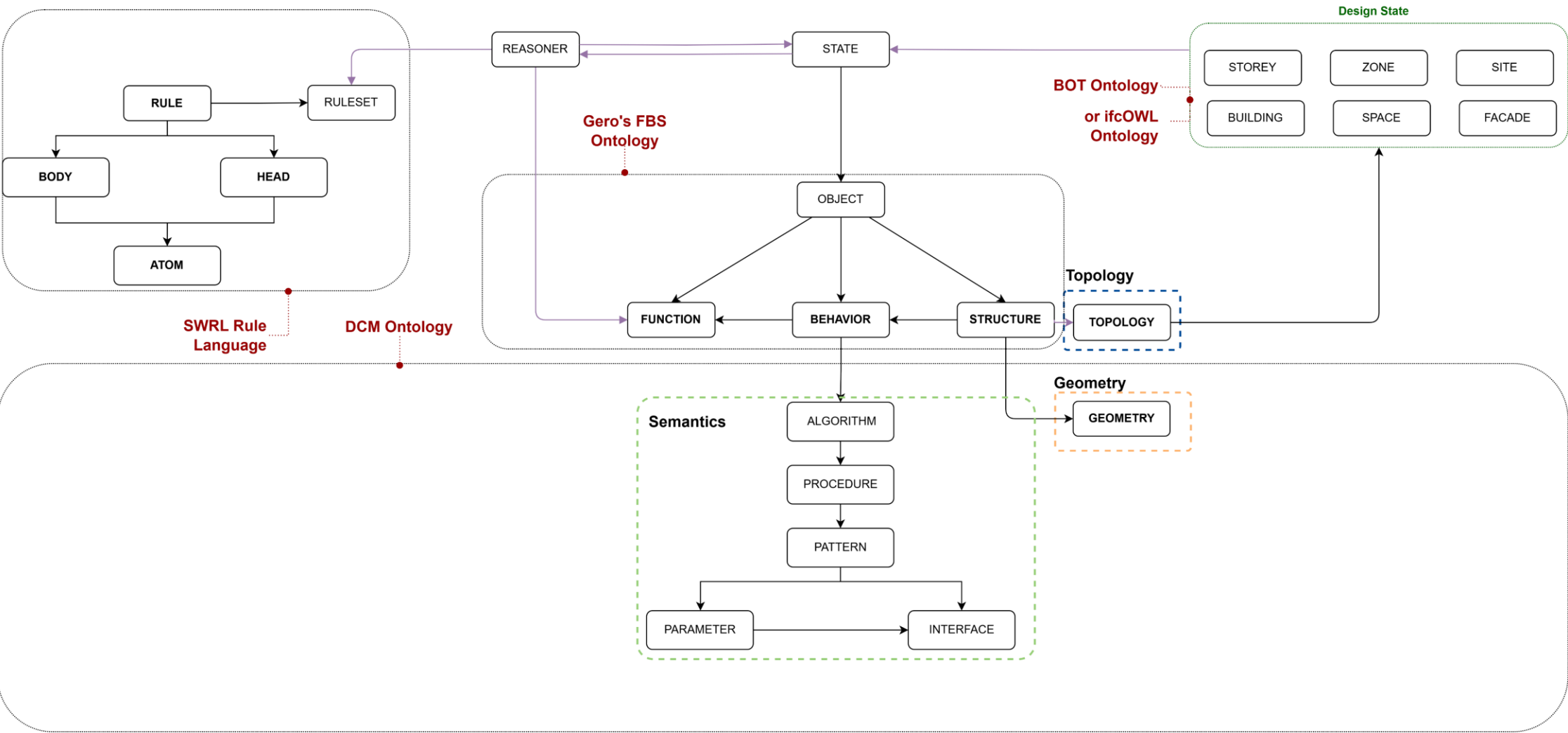
STEP 3: KEY CONCEPT MAP



3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 4: TAXONOMY OF CLASSES (CLASSES & HIERARCHY)

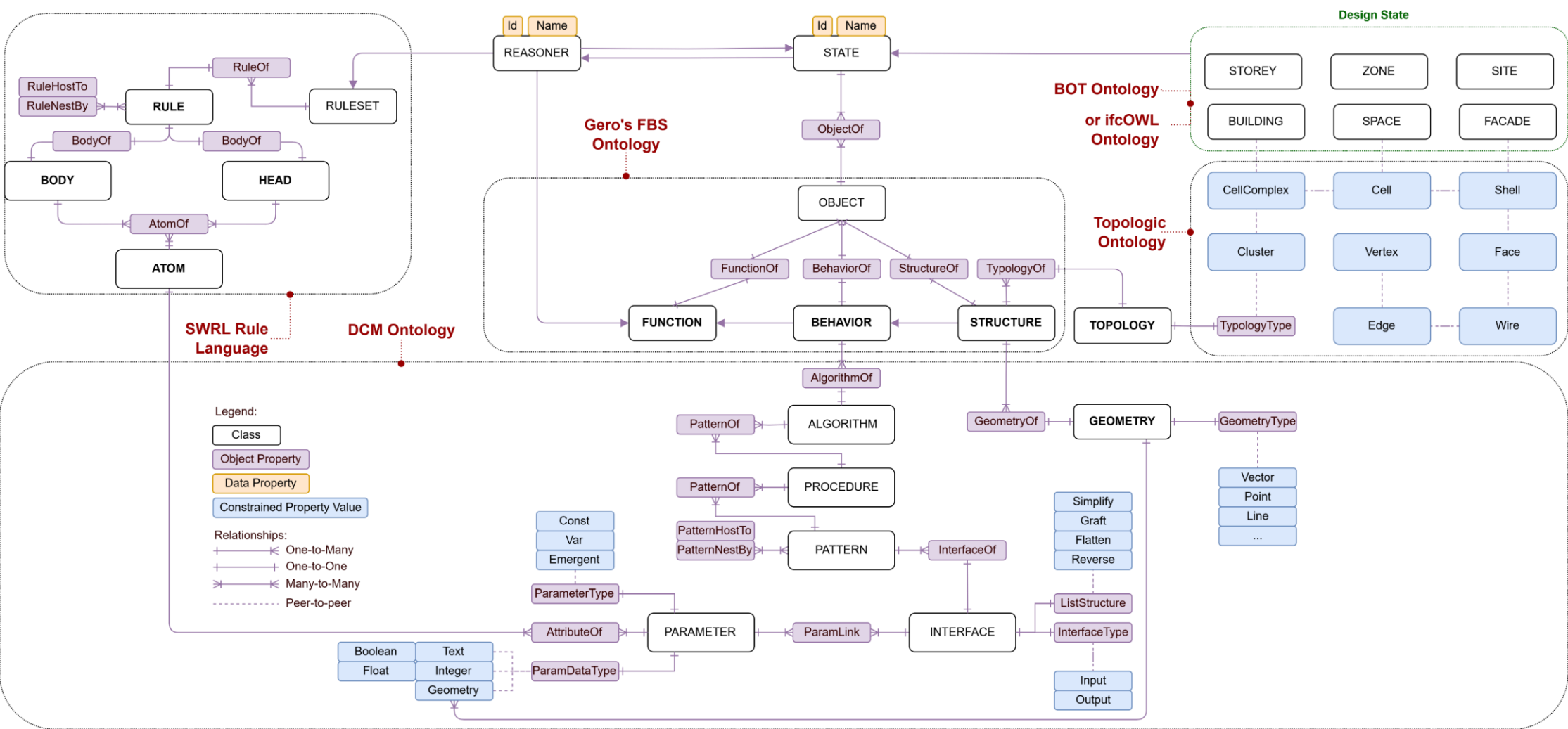
Entity-Relationship Diagram



3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 5: CLASS PROPERTIES – ATTRIBUTES & RELATIONSHIPS

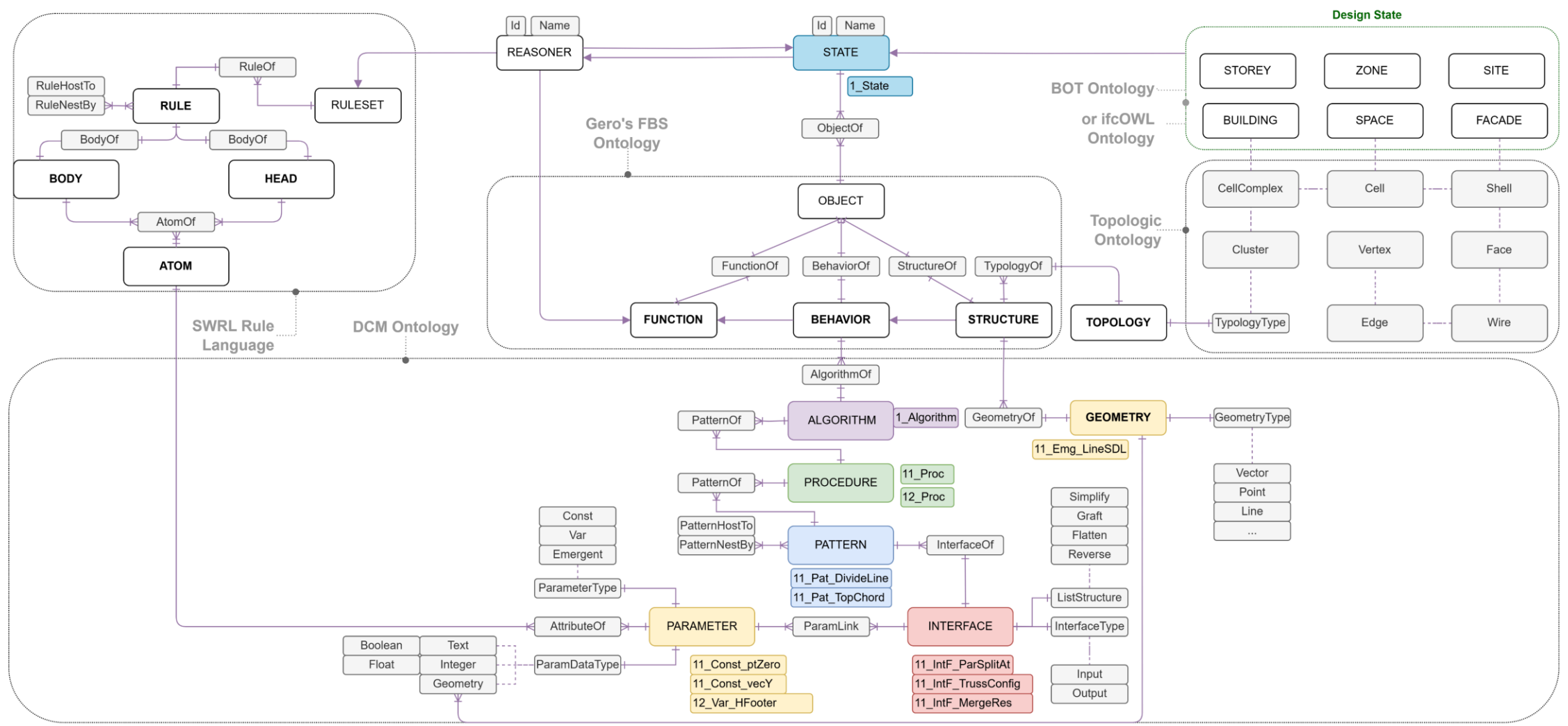
Entity-Relationship Diagram



3 ONTOLOGY DEVELOPMENT WORKFLOW

STEP 5: DEFINE PROPERTY CONSTRAINTS

Entity-Relationship Diagram



2 CONCEPT STAGE. ONTOLOGIES

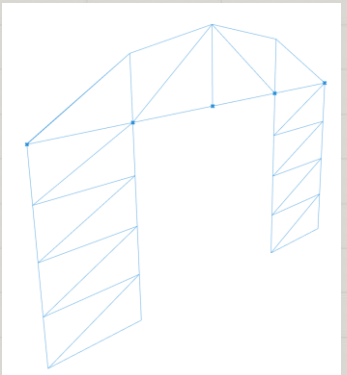
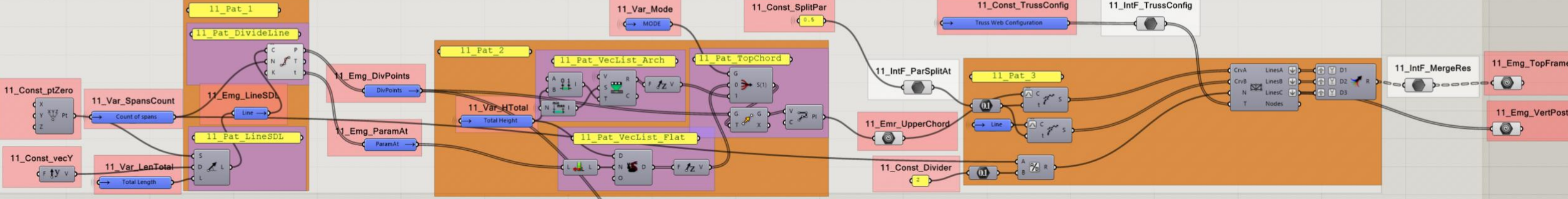
3 ONTOLOGY DEVELOPMENT WORKFLOW

PARAMETRIC DESIGN MAPPING TEST

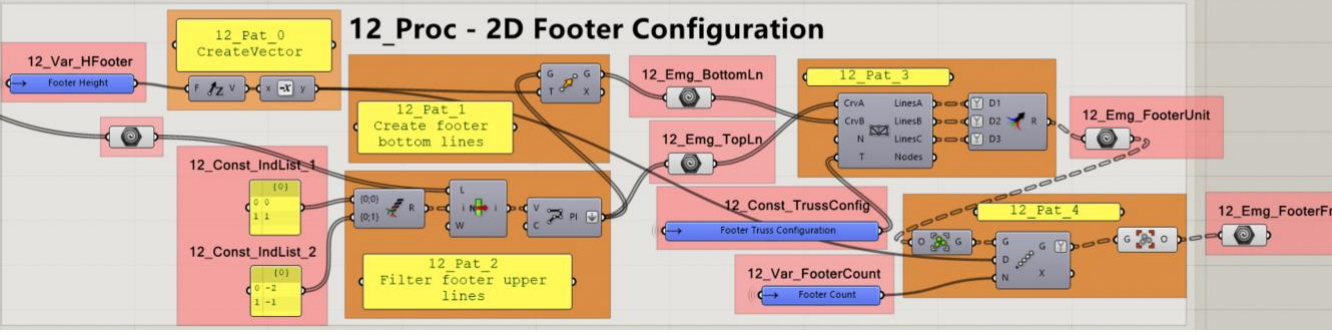
OBJECT - FRAME

1_ALGORITHM

11_Proc - 2D Truss Configuration



12_Proc - 2D Footer Configuration



2 CONCEPT STAGE. ONTOLOGIES

3 ONTOLOGY DEVELOPMENT WORKFLOW

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